**Challenge sponsored by** Verasonics®

# IEEE UFFC-JS 2024: a-MEM Challenge Instructions

## Aim

Participants are encouraged to process the supplied US imaging data sets to produce the best possible elastograms. Participants will need to characterise shear wave information from the data which will be representative of either an acoustic radiation force (ARF) impulse method, or a time harmonic vibration provided by external shakers. When producing elastograms and optimising their algorithms, participants will need to consider contrast to noise, spatial resolution and accuracy in spatial registration.

## Overview

Participants who wish to enter the challenge may do so by contacting the organising committee. Participants will be given access to a shared Dropbox folder from which to download the relevant data files in the form of '.mat' files (MATLAB). Each file will contain the raw data and relevant parameters associated with the acquisition of ultrasound imaging data that will enable detection of oscillations due to shear wave propagation in a phantom. The participants' aim is to generate high quality shear wave speed maps. (see below for information on how quality is assessed). Participants are free to use the available data in any way they deem appropriate to produce the best possible shear wave speed maps. To enter the challenge participants will be required to submit their results for either or both categories.

## Challenge rules:

Participants may enter the Challenge by contacting the organisers, giving contact details (name, email address, affiliation) via the registration form on the IEEE UFFC-JC website. There may be more than one entry from any given research team/department, however, the processing methods should be distinct, it is not enough to just vary the parameters of the same method. How they are distinct must be clear in the accompanying abstracts.

Upon acknowledgment by the organisers, participants will be given access to shared Dropbox folders from which to download the Challenge data.

There will be separate prizes for the two categories (ARF (impulse), harmonic), participants are therefore free to submit their results for either or both methods. For each category there are 3 imaging data sets. One data set is obtained from a uniform and homogeneous phantom material. Two further data sets are obtained from imaging a phantom with an inclusion within a homogenous background. Participants are free to experiment with the data and their algorithm settings, however in the interest of fairness we require that submitted entries are obtained using the same algorithm settings for all data sets within each category.

Participants are expected to supply a 2D shear wave speed map (please contact us if you wish to submit maps of elastic modulus) for each data set. These can be supplied directly to the organisers via email or other suitable file transfer methods.

Participants are expected to submit an abstract alongside their Challenge entry **following IEEE UFFC-JS conference guidelines** describing a summary of their methodology and their algorithm, however unlike the main conference abstracts these abstracts do not need to be blinded.

Below we will give details on how the quality of the elastograms will be evaluated. If you use these details to determine what your results might be, **please refrain from including them in the abstract.** You are welcome to include any figures that include other elastograms that you may have used to develop your method, but not those submitted to the Challenge.

Participants agree to present an oral summary of their work at the IEEE UFFC-JS symposium if they are invited to do so. As this is a hybrid conference this may be in-person or on-line.

Participants agree to share in confidence their code (or a compiled file if preferred) and instructions on how to apply it to the Challenge data, if requested.

The deadline for submission of data and accompanying abstracts is:

**June 10th 23.59 (Pacific Time).**

Any entries received after this date will not be considered.

A **joint publication** will be written after the challenge, for submission to IEEE UFFC. Please only enter the challenge if you agree to disclose your methods for the workshop and contribute details of your methods in your abstract, individual conference paper and this publication.

For use of the challenge data beyond IEEE UFFC-JS 2024, this publication must be cited.

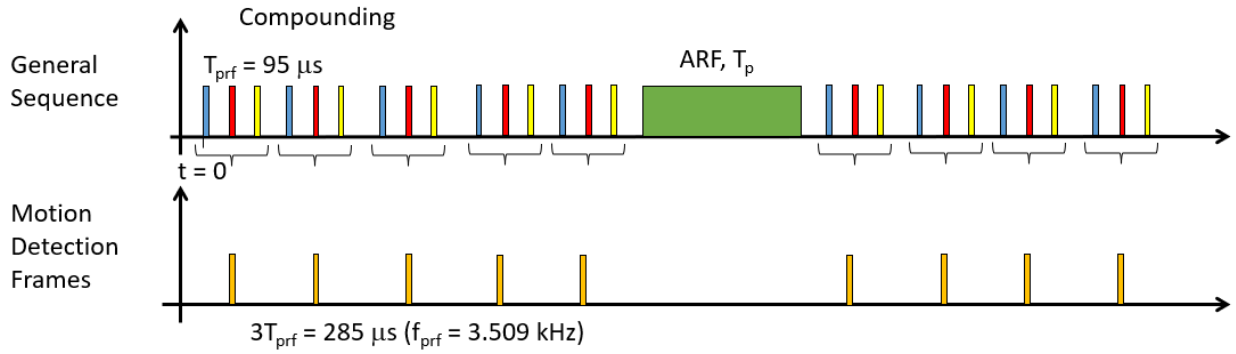
## ARF Method Data Acquisition

Data were taken with a L7-4 linear array transducer (Philips Healthcare, Andover, MA) and a Verasonics V1 system (Verasonics, Inc., Kirkland, WA). The push ultrasound frequency was 4.09 MHz and detection pulses were at 5.00 MHz. Radiofrequency (RF) data were sampled at 20.00 MHz. The demodulation frequency for the in-phase/quadrature (IQ) data was 5.00 MHz. Plane wave compounding was used with transmissions at -4°, 0°, +4°, and the pulse repetition period for each angle was 95 μs.

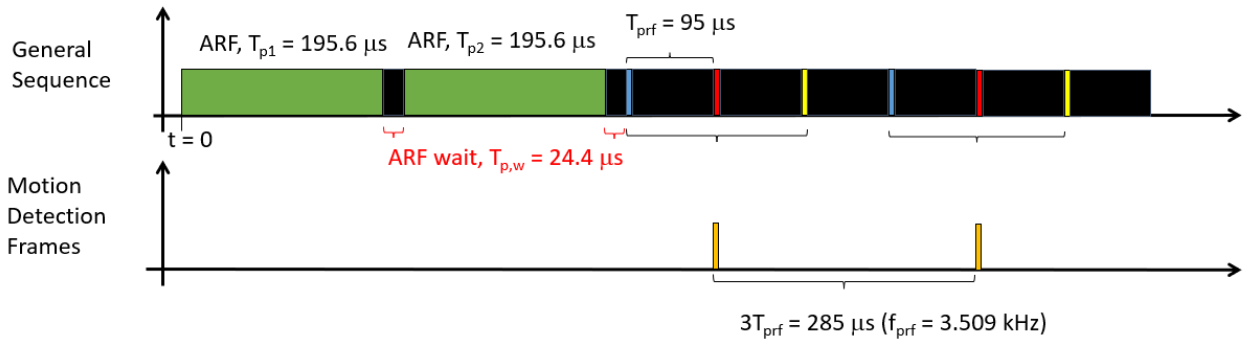
The compounding was performed in the Verasonics V1 system, and the compounded IQ data were provided as the output. An assumed speed of sound of 1540 m/s was used. The IQ data were reconstructed with $\Delta z = \Delta x = 0.5\lambda = 0.5 \cdot (1540 \text{ m/s})/(5 \text{ MHz}) = 0.154$ mm. The IQ data are complex data with size 325 x 247 x 110 (z, x, t).

The RF data are long vectors of data for each of the 128 elements in the array transducer. Each frame has a record length of 1536 samples. All the data are present for all three angled transmissions for the compounding. As a result, each vector is 1536 samples x 3 angles x 110 frames = 506,880 samples. The data are stored in the variable 'RF'. Additional structures from the V1 Verasonics acquisition are included in each RF file.

The sequence includes 5 compounded frames prior to the acoustic radiation force (ARF) push and 105 compounded frames after. The specific timings are given in the diagrams below.




- The blue, red, and yellow acquisitions represent different angles.
- Green represents ARF push transmission
- 5 motion detection sets taken before push and 105 motion detection sets taken after push




- The blue, red, and yellow acquisitions represent different angles.
- Green represents ARF push transmission
- Black represents waiting times

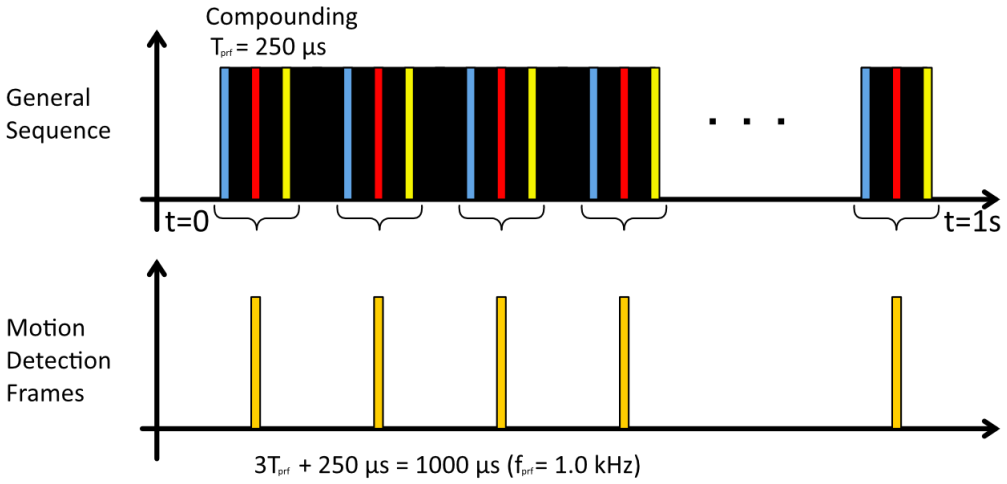
## Harmonic Method Data Acquisition

Data were taken with a C5-2v convex array transducer (Verasonics, Inc., Kirkland, WA) and a Verasonics Vantage 64 system (Verasonics, Inc., Kirkland, WA). The detection pulses were at 3.57 MHz. Radiofrequency (RF) data were sampled at 14.28 MHz. Plane wave compounding was used with transmissions at -12°, 0°, +12°, and the pulse repetition period for each angle was 250 μs. Between frames a waiting time of 500 μs was added, leading to an effective frame rate of 1000 Hz.

Harmonic multifrequency vibration at 100, 120, 140, 160, 180, 200 Hz were continuously applied during the entire measurement.

The compounding was performed in the Verasonics Vantage system, and the compounded IQ data were provided as the output. The IQ data were reconstructed with $\Delta z = \Delta x = \lambda$ = (1540 m/s)/(3.57 MHz) = 0.43137 mm. The IQ data are complex data with size 266 x 146 x 1002 (z, x, t).

The RF data are long vectors of data for each of the 64 center elements in the array transducer. Please note, that the transducer has 128 elements and only the center elements were used. Each frame has a record length of 8448 samples. All the data is present for all three angled transmissions for the compounding. As a result, each vector is 2816 samples x 3 angles x 1002 frames. The data are stored in the variable 'RF'. Additional structures from the acquisition are included in each RF file.




- The blue, red, and yellow acquisition represent different angles.
- Black represents waiting times

**Data Sets and Parameters**

The organisers provide separate '.mat' files each containing the raw data and information required to generate one 2D elastogram. The following files are provided for download:

| | |
|---|---|
| ARF_phantom_I.mat | ARF method phantom 1, homogeneous |
| ARF_phantom_II.mat | ARF method phantom 2 with inclusion |
| ARF_phantom_III.mat | ARF method phantom 3 with inclusion |
| Harmonic_phantom_I.mat | Harmonic method phantom 1, homogeneous |
| Harmonic_phantom_II.mat | Harmonic method phantom 2 with inclusion |
| Harmonic_phantom_III.mat | Harmonic method phantom 3 with inclusion |

Each file will contain the following variables, with associated units indicated in square brackets where necessary:

IQ_Data – In phase/Quadrature US imaging data in complex format (In phase real, Quadrature imaginary). For ARF methods multiple sets of ARF data may be provided representing data acquisition for different ARF push locations, these will appear as contents of separate cells, (IQ_Data{1}, IQ_Data{2},

etc…). Each data set will be a 3 dimensional array with 2 spatial dimensions (axial, lateral) and one temporal.

IQ_demodulation_f – The demodulation frequency used to generate the IQ data [MHz].

IQ_sample_rate – IQ data temporal sampling rate [MHz].

frame_rate – Processed IQ data frames rate [Hz].

SoS – Sound speed used for image and IQ data reconstruction [m/s].

vibration_frequency – Vibration frequencies used to generate shear waves in the harmonic method. A vector indicates multiple simultaneous harmonic frequency components [Hz].

X_IQ – Vector containing lateral direction coordinates for the provided IQ data [mm].

Z_IQ – Vector containing axial direction coordinates for the provided IQ data [mm].

dX – Step size in the lateral direction for the IQ data [mm].

dZ – Step size in the axial direction for the IQ data [mm].

X_push – Vector containing the lateral coordinates at which ARF pushes are centred. The number of elements will be equal to the number of data sets in IQ_data [mm].

**Access to Imaging RF Data**

Participants may wish to reconstruct their shear wave speed maps from the raw channel level digitised ultrasound echo data instead of the reconstructed IQ data. Participants who wish to use RF data should contact the organisers with a brief explanation of the reason for requiring access to the RF data. The organisers will then provide access to the RF data and associated parameters.

**Challenge Entries**

Participants can submit their entries directly to the organising committee via email (John.Civale01@icr.ac.uk) or other file transfer method (i.e. ZendTo). The preferred format is a MATLAB '.mat' file for each data set. Each file should contain a 2D shear wave speed map with the same spatial dimensions and resolution as the original IQ_data. Submissions should be saved in individual .mat files with a file name that is clearly associated with the filename of the original imaging data.

**Entry Assessment**

In the absence of an absolute ground truth as reference, entries will be assessed according to shear wave speed data map data (please contact us if you wish to submit maps of elastic modulus). For data sets without inclusions (i.e. homogeneous background) the spatial variation in shear wave speed will be measured across pre-defined ROIs. For data sets with an inclusion three criteria will be considered: contrast to noise ratio, edge sharpness, and edge accuracy. Contrast-to-noise ratio will be assessed by evaluating the properties of two predefined ROIs: the first will be a circularly shaped ROI that lies wholly within the inclusion, the second will be an annulus surrounding the inclusion but consisting of pixels wholly within the background region. Edge sharpness and accuracy will be assessed by considering the spatial gradient of shear wave speed in radial directions at multiple angles around the inclusion boundary. The spatial gradient (edge sharpness) and its distance from the boundary as determined from the B-mode image will provide relevant metrics. Results for all entries will be collated and ranked for each data set and assessment criteria, in total there will be 7 overall ranking criteria (2 data sets x 3 criteria for inclusions, + 1 data set x 1 criteria for homogenous) for each type of data category (ARF, harmonic). An overall winner will be declared for each category, by determining the entry with the lowest mean rank. In the event where entry data is missing (i.e. shear wave speed maps not submitted for one or more data sets) the worst possible rank equal to the total number of participants will be assigned. Similarly, we encourage participants to submit shear wave speed maps containing full numerical data (i.e. avoiding non-numeric value due to signal dropout), especially within and in the proximity of the assessment ROIs. If this cannot be avoided, the organisers will retain the right to apply a penalty or substitute numerical values of their choosing to data points missing numerical values. For transparency, we provide participants the parameters that will be used to define the circular and annular ROIs (phantoms with inclusions), and the rectangular ROI to measure variance in the homogeneous phantom data sets.

| | Centre | | ROI 1 | ROI 2 | |
|---|---|---|---|---|---|
| | X (mm) | Z (mm) | radius (mm) | r_inner (mm) | r_outer (mm) |
| ARF method phantom 2 with inclusion | 0.327 | 16.478 | 7.39 | 9.39 | 11.95 |
| ARF method phantom 3 with inclusion | 0.327 | 35.266 | 8.36 | 10.36 | 13.31 |
| Harmonic method phantom 2 with inclusion** | 1.050 | 43.607 | 6.81 | 9.81 | 11.94 |
| Harmonic method phantom 3 with inclusion | -0.245 | 31.959 | 7.42 | 10.42 | 12.79 |
| | | | | | |
| | X limits (mm) | | | Z limits (mm) | |
| ARF method phantom 1 homogeneous | -15.42 | 15.42 | | 4.00 | 46.51 |
| Harmonic method phantom 1 homogeneous* | -25.70 | 25.64 | | 19.88 | 75.96 |

N. B. There was a mistake in naming the **IQ data files** on the Dropbox folder, please note the following:

* The evaluation ROI definitions are for the data in filename:
*Harmonic_phantom_II.mat*

** The evaluation ROI definitions are for the data in filename:
*Harmonic_phantom_I.mat*

When providing results files, please retain file name nomenclature from the original supplied files.

**The IEEE UFFC-JS Challenge Team:**

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